

Types of Bi-clusters

- ❑ A bi-cluster: A submatrix where objects (e.g., genes) and attributes (e.g., conditions) follow some consistent patterns
- ❑ 4 types of bi-clusters (ideal cases)
 - ❑ Bi-clusters with constant values: for any i in I and j in J , $e_{ij} = c$
 - ❑ Bi-clusters with constant values on rows: $e_{ij} = c + \alpha_i$, can be constant values on columns
 - ❑ Bi-clusters with coherent values (aka. pattern-based clusters): $e_{ij} = c + \alpha_i + \beta_j$
 - ❑ Bi-clusters with coherent evolutions on rows: $(e_{i_1j_1} - e_{i_1j_2})(e_{i_2j_1} - e_{i_2j_2}) \geq 0$, i.e., only interested in the up- or down- regulated changes across genes or conditions without constraining on the exact values

10	10	10	10	10	10	50	30	70	20	10	50	30	70	20
20	20	20	20	20	20	60	40	80	30	20	100	50	1000	30
50	50	50	50	50	50	90	70	110	60	50	100	90	1200	80
0	0	0	0	0	0	40	20	60	10	0	80	20	100	10